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1 create database Bank;
2 create table Customer(CustomerID int Primary key,
3 CustomerName varchar(50),PhoneNo Varchar(25),City varchar(20),
4 AccountType varchar(10),AccountNo int);
5 --Data enter--
6 insert into Customer values
7 (1,'Rahul Sharma',9876543210,'Pune','Saving',1001),
8 (2,'Sneha Patil',9988776655,'Mumbai','Current',1002),
9 (3,'Aman Verma',9123456780,'Nagpur','Saving',1003),
10 (4,'Priya Singh',9012345678,'Delhi','Current',1004),
11 (5,'Karan Mehta',9871203456,'Hyderabad','Saving',1005),
12 (6,'Neha Joshi',9988001122,'Pune','Current',1006),
13 (7,'Rohit Kumar',9765432109,'Bangloare','Saving',1007),
14 (8,'Pooja Sharma',987654001,'Chennai','Saving',1008),
15 (9,'Vivek Shah',9001122334,'Ahmedabad','Current',1009),
16 (10,'Anjali verma',9988771100,'Jaipur','Saving',1010);
17 select * from customer;
18 -- Accounts Table---
19 CREATE TABLE Accounts (
20     AccountID Int,
21     CustomerID INT,
22     Balance INT,
23     OpenDate DATE
24 );
25
26 INSERT INTO Accounts(AccountID, CustomerID, Balance, OpenDate)
27 VALUES (1001, 1, 55000, '2025-01-10'),
28 (1002, 2, 120000, '2024-11-20'),
29 (1003, 3, 35000, '2025-03-15'),
30 (1004, 4, 98000, '2025-02-01'),
31 (1005, 5, 75000, '2025-01-25'),
32 (1006, 6, 150000, '2024-12-18'),
33 (1007, 7, 42000, '2025-04-10'),
34 (1008, 8, 88000, '2025-05-05'),
35 (1009, 9, 200000, '2024-09-30'),
36 (1010, 10, 67000, '2025-03-22');
37 select*from Accounts;
38
39 -- Transactions Tables --
40 Create Table [Transactions] (TransactionID Int,AccountID INT,
41 TransactionType Varchar(50),Amount Int,TransactionDate Date);
42 Insert into [Transactions] (TransactionID,AccountID,TransactionType,
43 Amount,TransactionDate)
44 Values(1,1001,'Deposit',10000,'2026-06-01'),
45 (2,1002,'Withdraw',5000,'2026-06-02'),
46 (3,1003,'Deposit',25000,'2026-06-02'),
47 (4,1003,'Withdraw',3000,'2026-06-03'),
48 (5,1004,'Deposit',15000,'2026-06-04'),
49 (6,1005,'Deposit',12000,'2026-06-05'),
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50 (7, 1006, 'Withdraw', 7000, '2026-06-05'),
51 (8, 1007, 'Deposit', 9000, '2026-06-06'),
52 (9, 1008, 'Withdraw', 4500, '2026-06-06'),
53 (10, 1009, 'Deposit', 30000, '2026-06-07'),
54 (11, 1010, 'Withdraw', 2000, '2026-06-07'),
55 (12, 1002, 'Withdraw', 10000, '2026-06-08'),
56 (13, 1003, 'Deposit', 5000, '2026-06-08'),
57 (14, 1005, 'Withdraw', 3500, '2026-06-09'),
58 (15, 1007, 'Deposit', 15000, '2026-06-09');
59 select*from[Transactions];
60
61 -- Loan Table --
62 CREATE TABLE Loan(
63     LoanID INT ,
64     CustomerID INT,
65     LoanAmount INT,
66     LoanType VARCHAR(50)
67 );
68 INSERT INTO Loan (LoanID, CustomerID, LoanAmount, LoanType)
69 VALUES
70 (1, 1, 500000, 'Home Loan'),
71 (2, 2, 200000, 'Car Loan'),
72 (3, 4, 100000, 'Education Loan'),
73 (4, 5, 300000, 'Business Loan'),
74 (5, 6, 150000, 'Personal Loan'),
75 (6, 8, 250000, 'Home Loan'),
76 (7, 9, 400000, 'Business Loan'),
77 (8, 10, 180000, 'Car Loan');
78 SELECT * FROM Loan;
79
80 --- Task ---
81 --1. Display customer names, account numbers, and account balances using INNER JOIN.
82 select c.CustomerName,
83 a.AccountID,
84 a.Balance
85 FROM Customers c
86 INNER JOIN Account a on c.CustomerID = a.CustomerID;
87 --2. Find the top 3 customers with the highest account balances.
88 SELECT *
89 FROM Customers
90 ORDER BY balance DESC;
91 --3. Show all customers who have taken loans along with loan amount and loan type
92 SELECT
93     c.CustomerName,
94     l.LoanAmount,
95     l.LoanType
96 FROM Customers c
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97 INNER JOIN Loans l
98 ON c.CustomerID = l.CustomerID;
99
100 --4.Find the total deposited amount and total withdraw amount separatelt.
101 SELECT TransactionType,
102         SUM(Amount) AS TotalAmount
103 FROM [Transaction]
104 GROUP BY TransactionType;
105
106 --5.Dispaly customer wise toatl transaction amount using group by.
107 select c.CustomerID,
108        c.Customername,
109        SUM(t.Amount)AS Total_transaction
110 from Customers c
111 join Account a
112 on c.CustomerID = a.CustomerID
113 join [Transaction]t
114 on a.AccountID = t.AccountID
115 Group by c.CustomerID,c.CustomerName;
116
117 --6.Find customer whose balance is greter than average bank balance
118 select c.CustomerID,
119        c.Customername,
120        a.Balance
121 from Customers c
122 join Account a
123 on c.CustomerID = a.CustomerID
124 where a.Balance>
125 (
126 select Avg(balance)
127 From Account
128 );
129 -- 7. Show the hightest Transaction amount performed by customer.
130 SELECT c.CustomerID, c.CustomerName,
131        MAX(t.Amount) AS HighestTransactionAmount
132 FROM Customers c
133 JOIN Account a
134     ON c.CustomerID = a.CustomerID
135 JOIN [Transactions] t
136     ON a.AccountID = t.AccountID
137 GROUP BY
138     c.CustomerID,
139     c.CustomerName;
140 --8 Display customer who have not take loan using left join
141 SELECT
142     c.CustomerID,
143     c.CustomerName
144 FROM Customers c
145 LEFT JOIN Loan l
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146     ON c.CustomerID = l.CustomerID
147 WHERE l.LoanID IS NULL;
148 -- 9 Find the total number of transaction by each customer
149 SELECT
150     c.CustomerID,
151     c.CustomerName
152 FROM Customers c
153 join Accounts a
154 on c. customerID = a.CustomerID
155 Group BY c.CustomerID , c.CustomerName;
156
157 -- 10 Rank customer based on theri account balances using Rank()
158 select c.CustomerID,c.CustomerName,a.balance,
159 RANK()over(order by a.balance DESC)AS rank_no
160 From customer c
161 Join Account a
162 on c.CustomerID = a.CustomerID;
163 -- 11 Dens Rank customer based on theri account balances using DENS Rank()
164 select c.CustomerID,c.CustomerName,a.balance,
165     dense_RANK()over(order by a.balance DESC)AS dense_no
166 From customer c
167 Join Account a
168 on c.CustomerID = a.CustomerID;
169 --12 show previous transaction amount usig lag function
170
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